Integration — Lesson 25 Test: Inverse Trig Antiderivatives

10 questions · show your work in the box · no calculator · exact values only — leave π, e, and √ in your answers · don't forget + C on indefinite integrals

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Evaluate exactly: (a) $\arctan 1$ (b) $\arcsin(1/2)$ (c) $\arctan 0$. 1 pt

2. Find $\int 1/(1+x^2)\,dx$. 1 pt

3. Find $\int 3/\sqrt{1-x^2}\,dx$. 1 pt

4. Compute exactly: $\int_1^{\sqrt{3}} 1/(1+x^2)\,dx$. (Both limits are on the arctan exact-values table.) 1 pt

5. Compute exactly: $\int_0^{(\sqrt{3}/2)} 1/\sqrt{(1-x^2)}\, dx$. 1 pt

6. Find $\int 1/(16+x^2)\, dx$. (Hint: $16 = a^2$ for which a?) 1 pt

7. Compute exactly: $\int_0^{\sqrt{2}} 1/\sqrt{(4-x^2)}\, dx$. 1 pt

8. Three lookalikes — for each, name the right tool (arctan fact / u-sub → ln / power rule) AND give the antiderivative: (a) $\int x/(1+x^2)\, dx$ (b) $\int 1/(1+x)^2\, dx$ (c) $\int 1/(1+x^2)\, dx$. 1 pt

9. Verify by differentiating (show the chain-rule steps) that $(1/3) \arctan(x/3)$ is an antiderivative of $1/(9+x^2)$. 1 pt

10. Show that $\int_0^1 8/(1+x^2)\, dx = 2\pi$ exactly, and explain in one sentence why a result like this lets a computer calculate the digits of π using only areas. 1 pt